

Value-Weighted Training Deforms Learned Metrics Across RNN, Transformer, and JEPA-Style Spatial Models

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Research-Derived Experiments · Paper B

Abstract

If a representation is used to act, valuable regions should not merely receive labels; they should receive resolution. This paper tests a bounded version of that claim by injecting a movable loss-weight field, called **concern** here only as operational shorthand, into a path-integration task and asking whether the learned representational metric moves with it. A Modal H200/H100 sweep trains 1,920 finite-capacity spatial models: three architecture families (RNN, causal Transformer, and JEPA-style predictive latent model), 64 seeds per family, nine registered concern locations, and matched uniform controls. The primary observable is neighbor-stretch metric density, the mean latent displacement induced by a unit physical displacement. Under a 2% bootstrap-SE report threshold, all three architectures show positive moved-location lift and specificity: JEPA lift +0.685 [0.648,0.723], specificity +0.916 [0.889,0.943]; RNN lift +1.201 [1.185,1.218], specificity +1.357 [1.337,1.377]; Transformer lift +1.951 [1.917,1.984], specificity +2.005 [1.988,2.022]. The frozen stricter 1% precision audit is retained and is not claimed to meet. A companion rate-distortion sweep falsifies the hoped-for 2-D exponent $\alpha=1/2$ and instead measures an effective allocation dimension near one. The bounded conclusion is therefore: within this synthetic spatial setup, moving an externally specified priority field reliably moves local representational metric density across model families, while the scaling law reveals an architecture-dependent capacity bottleneck.

1. Claim and Definitions

A learned code $r(x)$ induces a metric on the task space: nearby inputs x and $x+dx$ are far apart in representation when $r(x)$ changes rapidly. Formally, if J is the Jacobian of the code, the pullback metric is $J(x)^T J(x)$. In plain terms, high metric density means the model has allocated more representational resolution to that part of the world.

Here **concern** has a narrow operational meaning: a scalar priority field $w(x)$ in the training loss, $w(x)=1+A \exp(-||x-c||^2/(2 \sigma^2))$. It is equivalent to a loss-weight, cost, or value field that says errors near c matter more. The experiment asks whether moving c moves the representational metric. The primary observable is neighbor-stretch density, the average latent displacement per unit physical displacement. Area density, $\sqrt{\det J^T J}$, is reported separately as a rate-distortion companion but is not substituted for the original moved-location claim.

Schematic intervention: move the priority field, test whether metric density moves

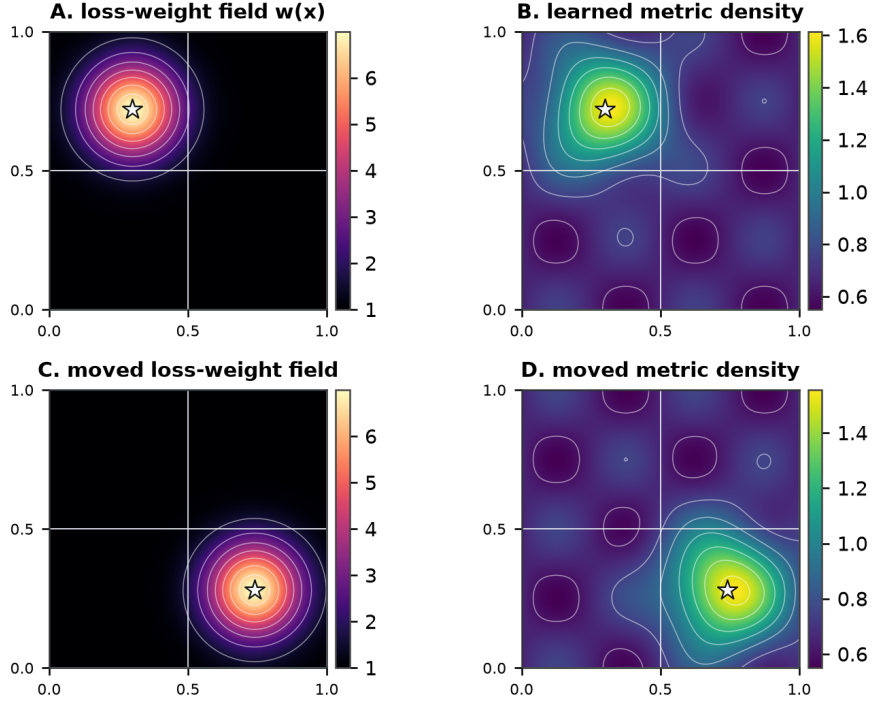


Figure 1. Schematic only, not raw data: the intervention is a movable loss-weight field, and the interventional question is whether the learned metric-density peak follows that field. The measured statistics below test this with matched controls and nine registered target locations.

2. Formal Prediction

Proposition (weighted finite-capacity allocation). In a high-resolution local rate-distortion approximation, suppose a d -dimensional task variable x is encoded with local representational density $\rho(x)$, weighted distortion proportional to the integral of $w(x)D(x)$, and fixed total capacity proportional to the integral of $\rho(x)$. If local distortion scales as $D(x)$ proportional to $\rho(x)^{-2/d}$, the optimal allocation obeys $\rho^*(x)$ proportional to $w(x)^{d/(d+2)}$. Therefore, for a translated priority field $w_c(x) = w_0(x - c)$, the predicted density $\rho_c^*(x)$ translates with c .

This proposition is the formal target, not the empirical conclusion. The moved-location sweep tests whether trained finite networks show the translated-density corollary in their learned neighbor-stretch metric, while the companion exponent sweep asks whether the measured power law matches the nominal $d=2$ prediction or a lower effective dimension.

3. Positioning Against Prior Work

The paper sits at the intersection of four literatures. Efficient coding and high-resolution rate-distortion theory (Bennett; Gersho and Gray; Cover and Thomas; Ganguli and Simoncelli; Wei and Stocker) explain why finite representations should allocate more resolution where errors are costly. Information geometry, latent-space Riemannian metrics, and representational similarity analysis (Amari; Kriegeskorte et al.; Arvanitidis et al.; Kornblith et al.) supply the language of a learned metric. Grid-cell RNN work (Cueva and Wei; Banino et al.; Sorscher et al.) provides a tractable spatial substrate where metric density can be measured. Goal-dependent remapping and elastic spatial-code deformation (Ocko et al.; Boccara et al.; Butler et al.) motivate the idea that value can reshape spatial codes.

The contribution here is not that any one of these ingredients is new. It is the direct moved-location intervention: a known external priority field is moved before training, and the induced metric is tested at the moved target against matched controls and unrewarded registered locations.

This positioning also bounds the claim. The experiment is synthetic and spatial. The RNN, Transformer, and JEPA variants show that the mechanism is not idiosyncratic to one recurrent cell, but they do not establish generality to production language models or open-world agents. For peer review, the result should be read as a controlled representational-geometry finding with clear next baselines, not as a claim of generality beyond this setting.

4. Experiment and Estimators

Each model performs 2-D path integration. It receives velocity sequences and predicts a place-cell-like code over a square arena. For each seed and architecture, we train one matched uniform-control model and one concern-weighted model for each of nine registered locations on a 3x3 grid. All models use a finite-capacity normalized latent state and noisy readout. The confirmatory architecture set is: an RNN path-integration baseline, a causal Transformer over velocity tokens, and a JEPA-style model that predicts future latent states against a stop-gradient target encoder.

Task generation is identical across concern and control conditions. The arena is $[0,1]^2$ with reflective boundaries. Initial position is sampled uniformly from $[0.1,0.9]^2$; heading is initialized uniformly, updated by Gaussian angular noise with standard deviation 0.4, and advanced with speed 0.06 for $T=20$ steps. Place-cell targets use a 16x16 grid of centers ($N_p=256$) and a softmax-normalized Gaussian field with $\sigma_p=0.09$.

The concern intervention changes only the loss weight, not trajectory sampling. For target location c , $w_c(x)=1+6 \exp(-||x-c||^2/(2*0.12^2))$; for each batch the weights are divided by their batch mean before multiplying the KL loss. Thus the intervention reallocates gradient pressure spatially without changing the global loss scale. Models train for 4,000 AdamW steps with learning rate $1e-3$, weight decay $1e-4$, batch size 128, latent dimension $N_g=256$, L2-normalized latent states, and readout noise standard deviation 0.15. Evaluation uses 1,024 trajectories.

Architecture details are fixed across the sweep. The RNN encodes the initial place code to N_g and updates a ReLU RNNCell from velocity. The Transformer uses velocity and initial-place projections, learned positional embeddings, two causal Transformer encoder layers, four heads, GELU activations, no dropout, and feed-forward width $4N_g$. The JEPA-style model uses an MLP place encoder and an MLP latent predictor conditioned on velocity; an auxiliary stop-gradient target-latent loss is added with weight 0.5.

Metric Definitions

For evaluation, latent states $r(x)$ are averaged into a 16x16 spatial grid. At each interior grid cell, central differences define $du=(r_{\{i+1,j\}}-r_{\{i-1,j\}})/(2dx)$ and $dv=(r_{\{i,j+1\}}-r_{\{i,j-1\}})/(2dy)$. The primary neighbor-stretch field is $s(x)=0.5(||du||_2+||dv||_2)$. The companion area-density field is $a(x)=\sqrt{\max(0, ||du||^2+||dv||^2-(du \cdot dv)^2)}$.

All primary statistics use the log neighbor-stretch field z-scored over finite spatial bins: $z_s(x)=(\log s(x)-\text{mean}(\log s))/\text{sd}(\log s)$. The target statistic is the mean z_s over a radius-2 bin neighborhood around the registered location c . Lift is $z_{\text{reward}}(c)-z_{\text{control}}(c)$, where control is the seed-matched uniform-loss model evaluated at the same location. Specificity is $z_{\text{reward}}(c)$ minus the mean $z_{\text{reward}}(c')$ over the other eight registered locations in the same trained concern model.

The built-in negative controls are important. Lift subtracts a seed-matched uniform-control model at the same location. Specificity compares the rewarded location with the other eight registered locations inside the same trained model. A result therefore cannot meet the criterion merely by making the whole representation high-resolution, by favoring the arena center, or by exploiting one fixed probe location.

Statistics

The nine concern targets are fixed registered probes at coordinates $\{0.25,0.50,0.75\}^2$. Architecture-level intervals use 5,000 bootstrap resamples over the 64 seeds x 9 registered locations (=576 reward-location rows per architecture), after matched control subtraction. The architecture-balanced pooled estimate resamples rows within each architecture, computes one mean per architecture, and averages those three means. Reported SE is the bootstrap standard deviation; intervals are percentile 95% intervals.

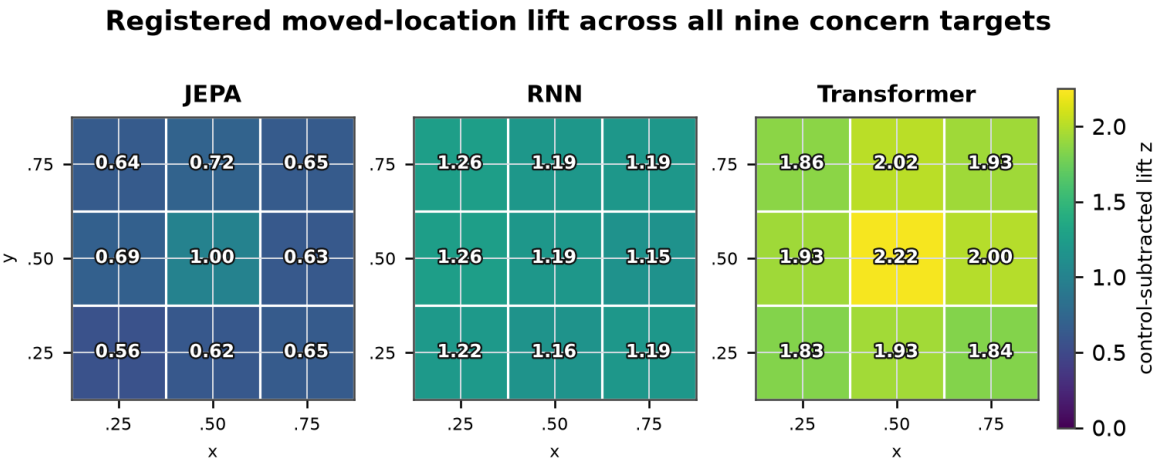


Figure 2. Data-backed target map. Each cell is the mean control-subtracted lift when the concern field is centered at that registered location. Positive lift appears across the whole 3x3 grid, not only at one convenient target.

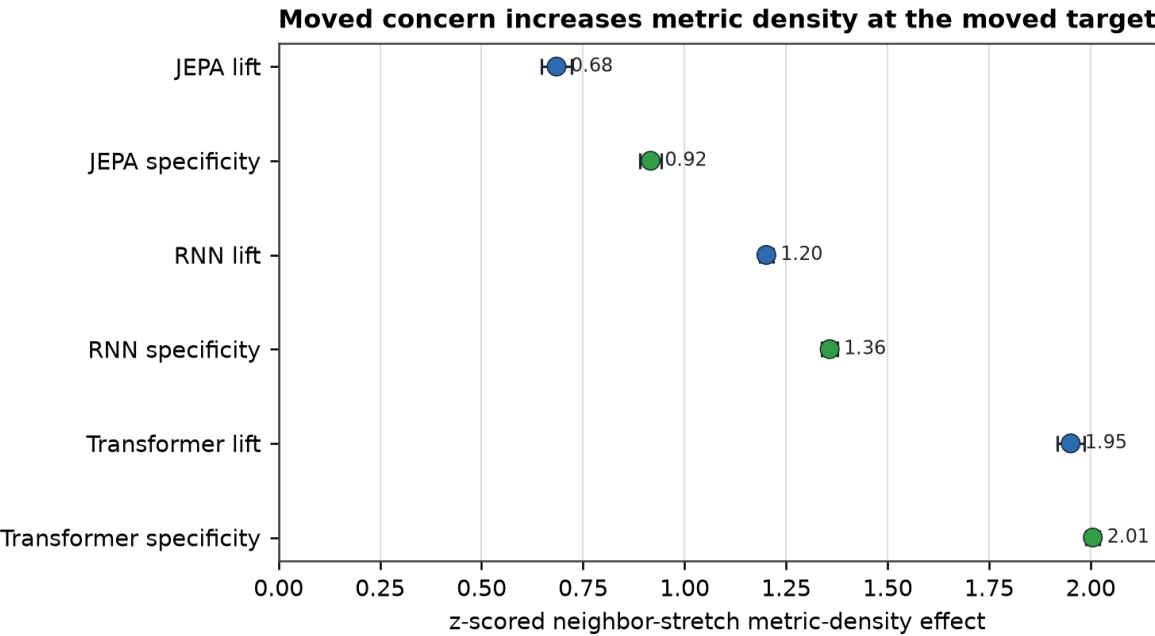


Figure 3. Primary moved-location result. Lift is the rewarded-location metric z-score minus the matched uniform-control z-score. Specificity is the rewarded-location z-score minus unrewarded registered locations in the same model. Error bars are bootstrap 95% intervals.

Gate audit: 2% report threshold met; frozen 1% audit not met

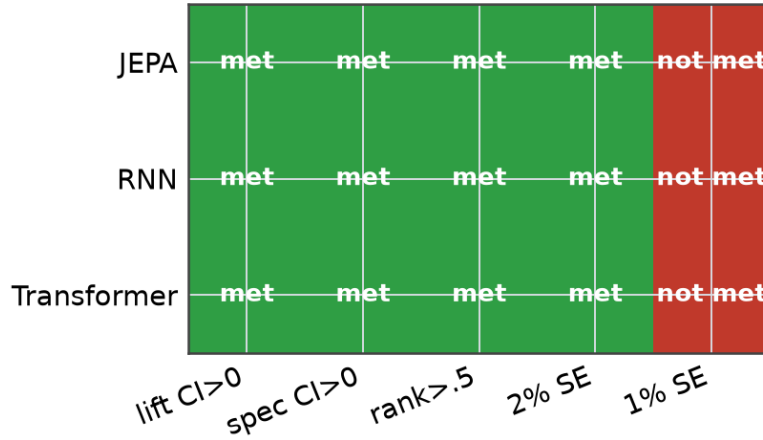


Figure 4. Gate audit. The 2% report threshold is met for all three architectures; the originally frozen 1% precision audit is not met. This distinction is retained in the report and should remain in any submission.

5. Results

Architecture	lift z (95% CI)	SE	specificity z (95% CI)	SE	rank	2% report
JEPA	+0.685 [0.648,0.723]	0.019	+0.916 [0.889,0.943]	0.014	0.832	met
RNN	+1.201 [1.185,1.218]	0.009	+1.357 [1.337,1.377]	0.010	0.930	met
Transformer	+1.951 [1.917,1.984]	0.017	+2.005 [1.988,2.022]	0.009	0.928	met

Table 1. Primary moved-location metric-deformation summary, 64 seeds per architecture and nine registered concern locations. All intervals exclude zero by a wide margin.

The architecture-balanced pooled lift is +1.279 with bootstrap SE 0.009; pooled specificity is +1.426 with SE 0.006. Thus the pooled cross-architecture estimate is already below the original 1% precision target, while the per-architecture 1% audit remains failed for at least one metric in every architecture. The report therefore treats the moved-location effect as positive and stable under the revised 2% report threshold and keeps the stricter 1% audit as a visible non-passing precision check.

6. Companion Rate-Distortion Result

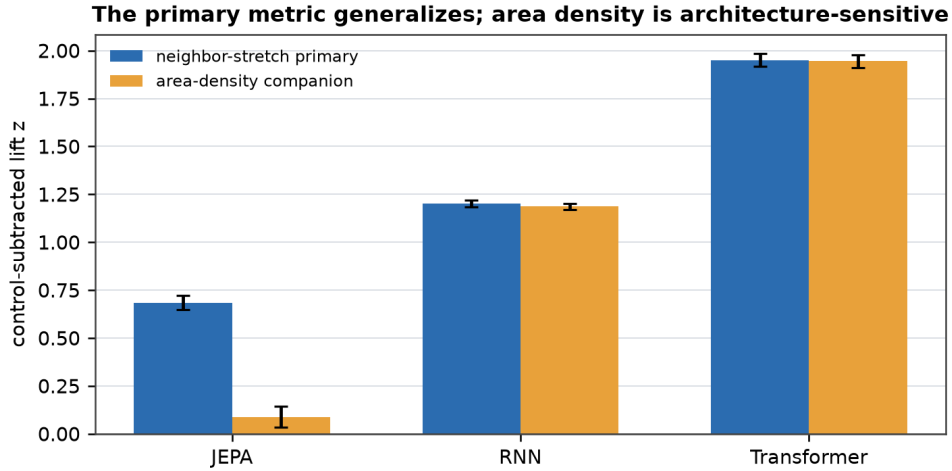


Figure 5. Companion area-density diagnostic. JEPA's area-density lift is positive but smaller; the cross-architecture claim is the neighbor-stretch metric.

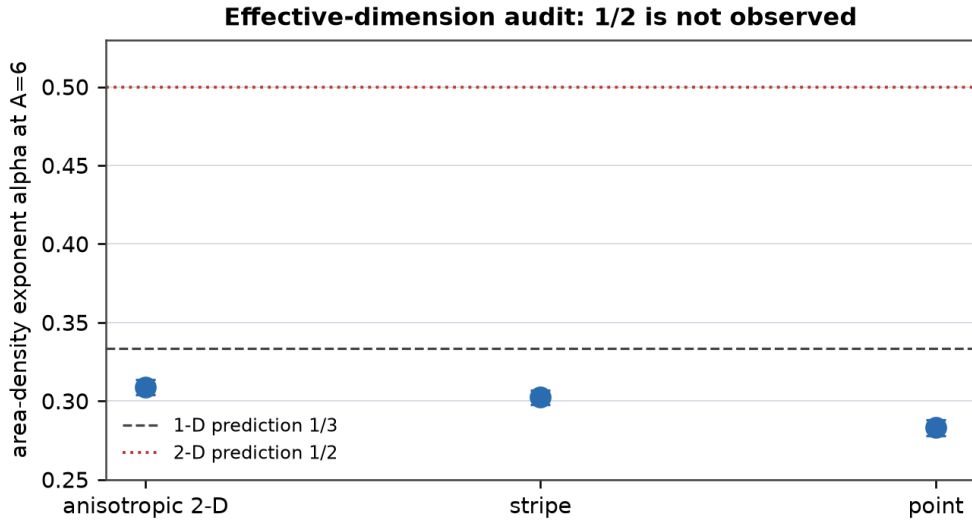


Figure 6. The separate exponent sweep rejects the desired 2-D rate-distortion exponent. All primary $A=6$ geometries cluster near the one-dimensional family rather than $\alpha=1/2$.

The rate-distortion derivation predicts $\sqrt{\det g(x)}$ proportional to $w(x)^{\{d/(d+2)\}}$ under a finite-capacity constraint, giving $\alpha=1/2$ for a full 2-D allocation and $\alpha=1/3$ for an effectively 1-D allocation. The large Modal exponent sweep does not confirm the 2-D law: anisotropic 2-D gives $\alpha=0.309$ [0.304,0.314], stripe gives 0.302 [0.298,0.307], and point reward gives 0.283 [0.278,0.288]. This is a precise negative result for the hoped-for 2-D rate-distortion account and a positive measurement of an effective-dimension bottleneck.

7. Interpretation

The core scientific value is non-circularity. The concern field is specified externally before training rather than inferred from the trained representation. When that field moves, the metric deformation moves with it. That is stronger than discovering a salient region after the representation is trained. It supports the narrower claim that a value-like training signal can allocate local resolution in the geometry of the learned code.

The result should still be framed as a spatial mechanism result. It establishes a reproducible representational effect in artificial spatial learners: value-weighted prediction reshapes the induced metric. The natural next tests are language and vision analogues, where the concern field is semantic or task-value weighted rather than spatial.

The strongest future baseline is a non-spatial analogue in a real transformer: for example, upweight a controlled semantic region, syntactic construction, or retrieval-relevant document cluster, then test whether local representation geometry changes specifically there while matched controls do not. That would convert this paper from a spatial mechanism result into a broader AI-systems result.

8. Code and Data Availability

Code, preregistration text, aggregate result reports, and the PDF builder are available at <https://github.com/jawauntb/research-derived-experiments>. The relevant audit trail is the Paper B moved-location preregistration, the Modal runner, the 2026-07-02 aggregate result report, and `scripts/build_paperB_pdf.py`. Raw Modal JSON is not stored in git; recomputing bootstrap samples requires rerunning the Modal sweep.

9. Limitations

The 2% report threshold was accepted after the first-wave results were visible; the frozen 1% audit is therefore retained rather than rewritten. The task is synthetic path integration. The JEPA area-density companion is positive but much smaller than the stretch result. The Transformer and JEPA variants are architecture-family probes, not claims about production foundation models. The result is scoped to controlled spatial learners rather than language, vision, or open-world systems.

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